

Power BI DAX – Advanced Concepts.

Take Your DAX Skills to the Next Level

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OVERVIEW :

Topic :

1. **ALL() Function – Remove Filters**
2. **REMOVEFILTERS() vs ALL()**
3. **SELECTEDVALUE() Function**
4. **RANKX() – Ranking Data**
5. **FILTER() Function**
6. **Time Intelligence – SAMEPERIODLASTYEAR()**
7. **DATESYTD(), DATESMTD(), DATESQTD()**



ALL() Function – Remove Filters :

- Removes filters from columns or tables
- Helps in calculating grand totals or percentage of total



Example :

Total All Sales = CALCULATE([Total Sales], ALL(Sales))

REMOVEFILTERS() vs ALL():

- **REMOVEFILTERS()**: Clears filters without changing shape of data
- **ALL()**: Removes filters AND returns all rows (can change row context)
- Use **REMOVEFILTERS()** for cleaner control of filter context



Example :

Selected Region = SELECTEDVALUE(Region[Name], "All Regions")

SELECTEDVALUE() Function :

- Returns a single value when only one value is selected, otherwise blank
- Great for dynamic titles or labels



Example :

Selected Region = SELECTEDVALUE(Region[Name], "All Regions")

RANKX() – Ranking Data :

- Ranks values in a table based on a measure .



Example :

Sales Rank = RANKX(ALL(Customer[Name]), [Total Sales])

FILTER() Function :

- Creates a table expression with rows that meet specific conditions
- Used inside CALCULATE()



Example :

```
CALCULATE([Total Sales], FILTER(Sales, Sales[Amount] > 1000))
```

Time Intelligence – **SAMEPERIODLASTYEAR()** :

- Used for year-over-year comparison



Example :

```
Sales LY = CALCULATE([Total Sales],  
SAMEPERIODLASTYEAR(Date[Date]))
```


DATESYTD(), DATESMTD(), DATESQTD() :

- Functions for calculating running totals



Example :

Sales YTD = TOTALYTD([Total Sales], Date[Date])

Sales QTD = TOTALQTD([Total Sales], Date[Date])